

To Bee or Not To Bee

Machine Learning Report - IG.2412

Date: 2026-06-06

This report summarizes the feature extraction, visualization, modeling, class imbalance experiments, and final model selection for pollinator insect classification.

Executive Summary

The best accuracy-oriented model is a soft voting ensemble combining SVM, shrinkage LDA, and ExtraTrees. It achieved 5-fold CV accuracy 0.8024 +/- 0.0462 and holdout accuracy 0.7600. A balanced-class alternative was also saved for minority-class-sensitive scoring.

Dataset

The training set contains 250 labeled samples with masks. Images 251-347 are the hidden test set described in the project brief, but the current workspace does not contain those test images.

Raw class distribution

bug type	count
Bee	115
Bumblebee	100
Butterfly	15
Hover fly	9
Wasp	9
Dragonfly	1
Bee & Bumblebee	1

Dragonfly and Bee & Bumblebee each have one sample. They were excluded from stratified validation, leaving the five evaluated classes below.

bug type	count
Bee	115
Bumblebee	100
Butterfly	15
Hover fly	9
Wasp	9

Top species labels

species	count
Bombus hortorum	71
Apis mellifera	58
Bombus pascuorum	25
Anthidium manicatum	19
Megachile centuncularis	17
Eristalis	9
Vespula germanica	8
Anthidium	6
Macroglossum stellatarum	5
Andrenidae	4

Feature Extraction

Symmetry: horizontal and vertical mask overlap plus mask aspect ratio.

Shape: Hu moments, solidity, eccentricity, extent, perimeter, area, convex area, and compactness.

Color: RGB min, max, mean, median, and standard deviation inside the insect mask.

Size/background: bug pixel ratio and background brightness.

Texture/appearance: color dominance, edge density, texture contrast, brightness mean, and saturation mean.

Engineered additions: RGB ranges, channel ratios, yellow/darkness proxies, bug-background brightness differences, symmetry interactions, shape ratios, Hu aggregates, and edge-texture products.

Dataset Imbalance: Majority Classes Dominate the Training Set

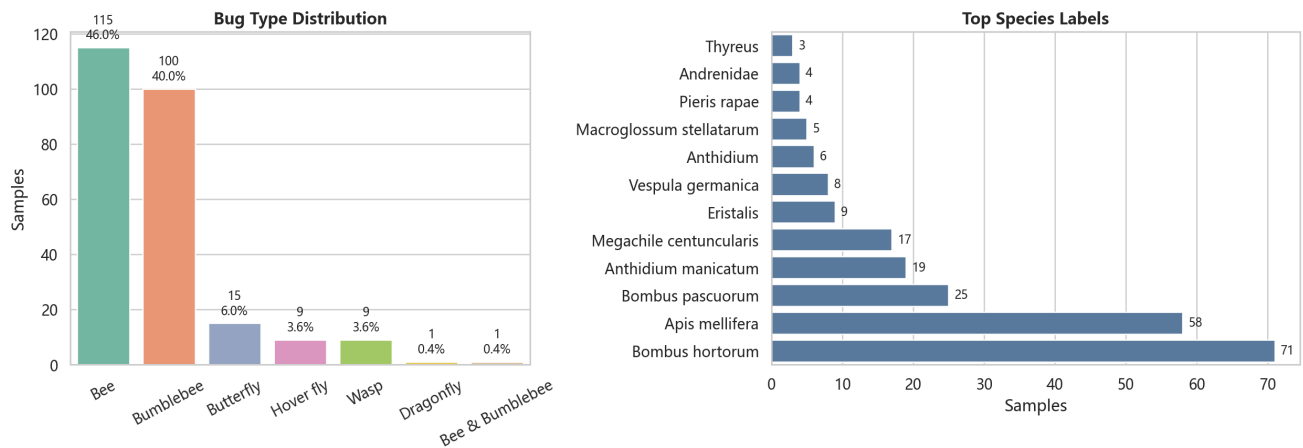


Figure 1. Annotated bug type imbalance and top species labels.

Visual Exploration

PCA, t-SNE, and UMAP show partial separation, but the dominant Bee and Bumblebee classes overlap. This overlap explains why model fusion and richer color/shape interactions help more than a single model.

Projection Comparison on 80 Engineered Features

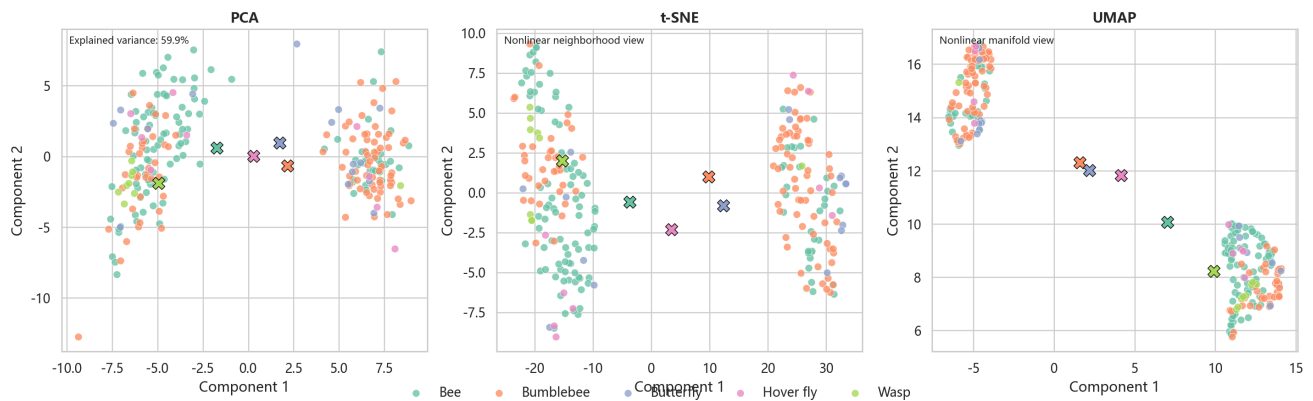


Figure 2. PCA, t-SNE, and UMAP comparison on engineered features.

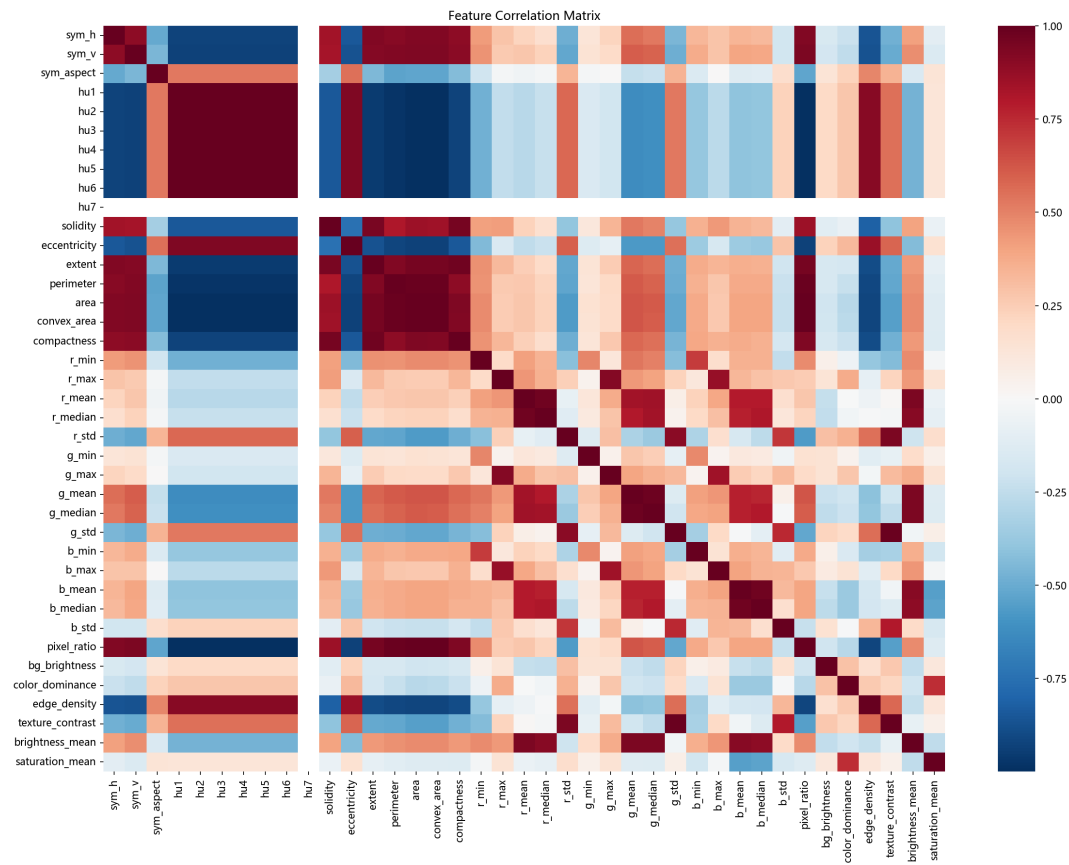


Figure 3. Feature correlation heatmap.

Modeling Protocol

The corrected workflow places StandardScaler inside each sklearn Pipeline, preventing test folds from influencing scaling parameters. Evaluation uses stratified 5-fold cross-validation and a fixed 80/20 holdout split.

Initial notebook models and clustering

Model	Score
Logistic Regression	0.7094+/-0.0504
KNN (k=5)	0.7220+/-0.0449
SVM (RBF)	0.7178+/-0.0415
Random Forest	0.7253+/-0.0654
KMeans	Sil=0.1876 ARI=0.0808
DBSCAN	Sil=0.6111 ARI=0.0814
LDA	0.7420+/-0.0391

KMeans and DBSCAN produced low adjusted Rand index values, so unsupervised clusters do not directly recover the target classes in the hand-crafted feature space.

Tuned supervised comparison

Model	CV Mean	CV Std	Test Acc
Logistic Regression (balanced)	0.6407	0.0725	0.6800
KNN (best)	0.7382	0.0524	0.6000
SVM (best)	0.7623	0.0436	0.7000
Random Forest (best)	0.7053	0.0594	0.7000
Gradient Boosting	0.6850	0.0683	0.6400
LDA	0.7420	0.0391	0.7000

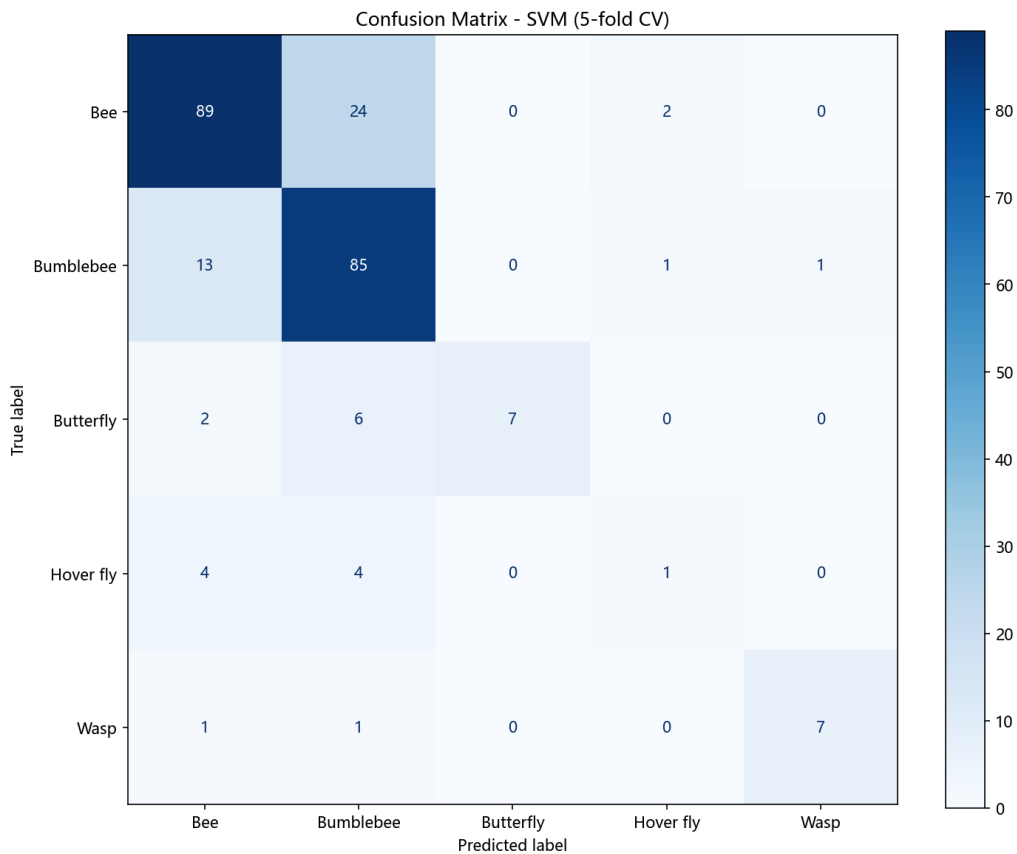


Figure 4. Cross-validated confusion matrix for the tuned SVM baseline.

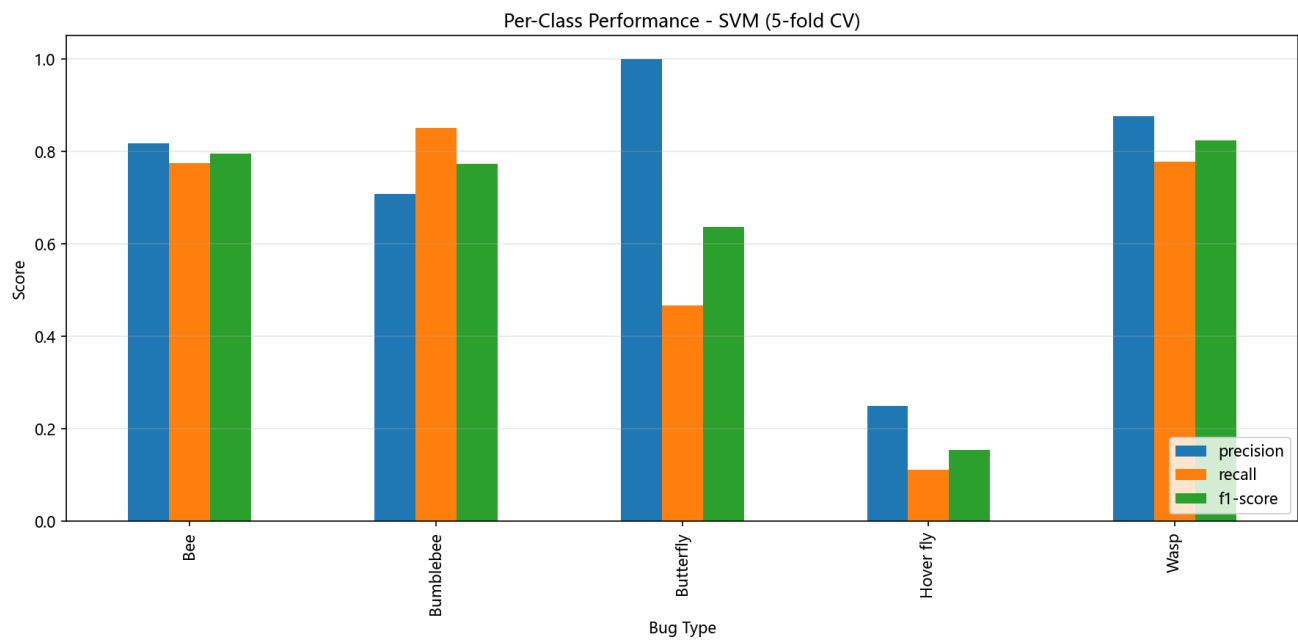


Figure 5. Per-class performance for the tuned baseline.

Corrected Validation and Feature Engineering

After fixing scaler placement, the original-feature SVM CV accuracy was 0.7504. With engineered features, SVM reached 0.7702. The final voting ensemble reached 0.8024, showing that feature engineering and model fusion both helped.

Class Imbalance and Advanced Models

The sweep tested no resampling, equal oversampling, hybrid_50 resampling, and mild undersampling. It also compared SVM variants, Logistic Regression, LDA/QDA, KNN, Naive Bayes, Random Forest, ExtraTrees, Gradient Boosting, HistGradientBoosting, AdaBoost, Voting, and Stacking.

Model	Imbalance Strategy	CV Accuracy Mean	CV Macro F1 Mean	Holdout Accuracy	Holdout Macro F1
Voting SVM+LDA+ET	none	0.8024	0.5979	0.7600	0.6040
SVM RBF tuned gamma=0.01	none	0.7744	0.5876	0.7200	0.4860
SVM RBF C=10	none	0.7702	0.5783	0.7000	0.4771
LDA shrinkage	none	0.7660	0.6071	0.7200	0.6840
SVM RBF C=10 balanced	none	0.7660	0.5630	0.7400	0.6058
ExtraTrees balanced	none	0.7540	0.5019	0.6800	0.2958
Voting SVM+LDA+ET	hybrid_50	0.7500	0.6037	0.7400	0.6957
Random Forest balanced	none	0.7456	0.5112	0.7000	0.3011
HistGradientBoosting	none	0.7420	0.5080	0.7600	0.4536
Voting SVM+LDA+ET	undersample_mild	0.7379	0.5927	0.7000	0.6256

XGBoost, LightGBM, and CatBoost

Additional boosted-tree libraries were tested on the same 80 engineered features. The best boosted-tree model did not exceed the soft voting ensemble, so the final accuracy-oriented recommendation remains unchanged.

Model	Family	Imbalance Strategy	CV Accuracy Mean	CV Macro F1 Mean	Holdout Accuracy
CatBoost compact	catboost	balanced_weight	0.7620	0.6158	0.6400
XGBoost conservative	xgboost	none	0.7537	0.5069	0.7000
LightGBM shallow	lightgbm	balanced_weight	0.7458	0.5473	0.6800
CatBoost compact	catboost	none	0.7457	0.4387	0.7000
LightGBM shallow	lightgbm	none	0.7456	0.5190	0.7000
XGBoost deeper	xgboost	none	0.7456	0.4735	0.7200
LightGBM compact	lightgbm	balanced_weight	0.7418	0.5345	0.6800
LightGBM compact	lightgbm	none	0.7418	0.4904	0.6800

Best boosted-tree CV accuracy: CatBoost compact / balanced_weight at 0.7620. Best boosted-tree macro F1: CatBoost shallow / balanced_weight at 0.6172, but its holdout macro F1 was 0.3208.

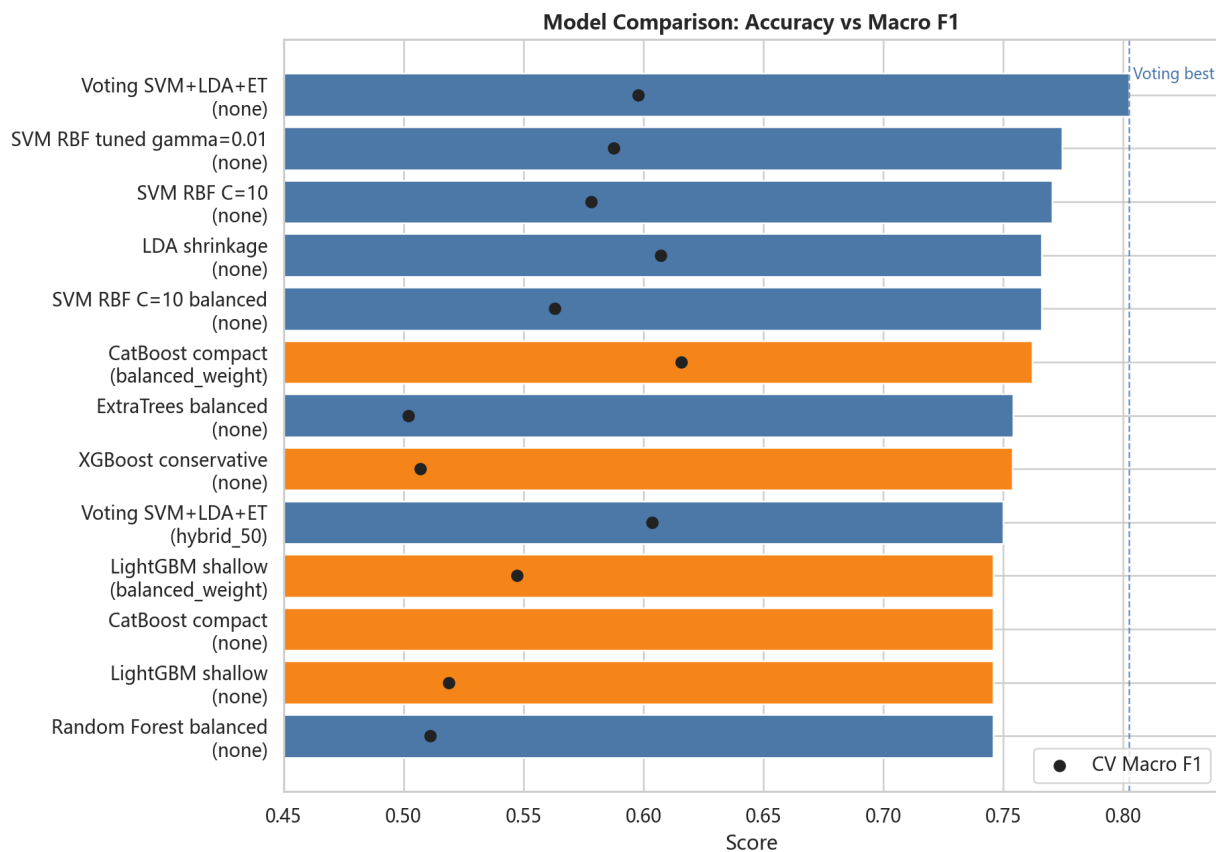


Figure 6. Final model comparison across fusion and boosted-tree approaches.

Holdout Confusion Matrix: Accuracy-Oriented Final Model

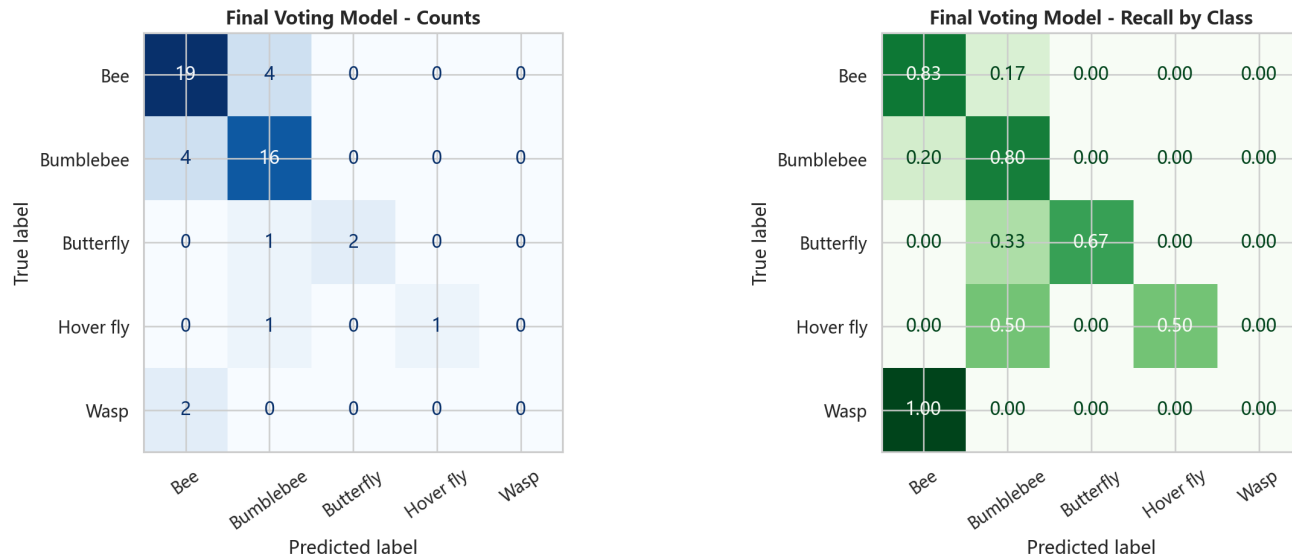


Figure 7. Holdout confusion matrix for the final voting model.

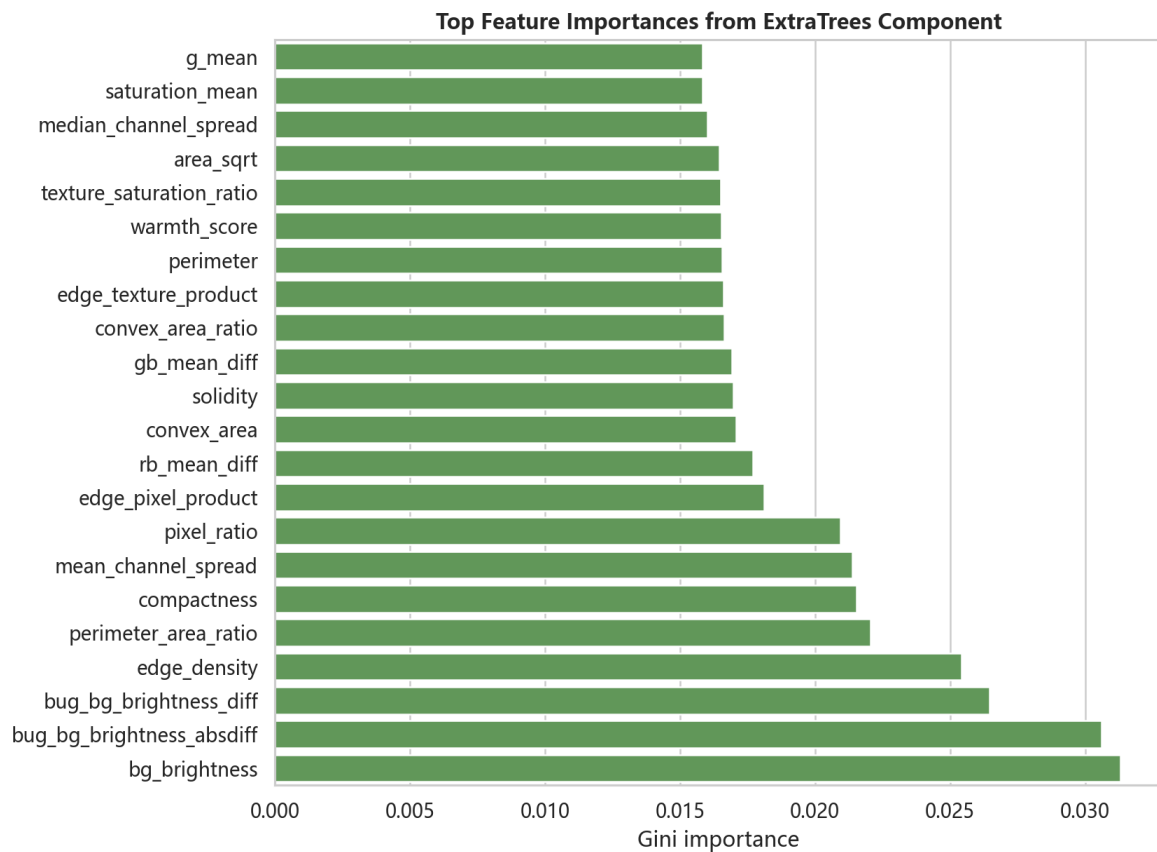


Figure 8. Top feature importances from the ExtraTrees component.

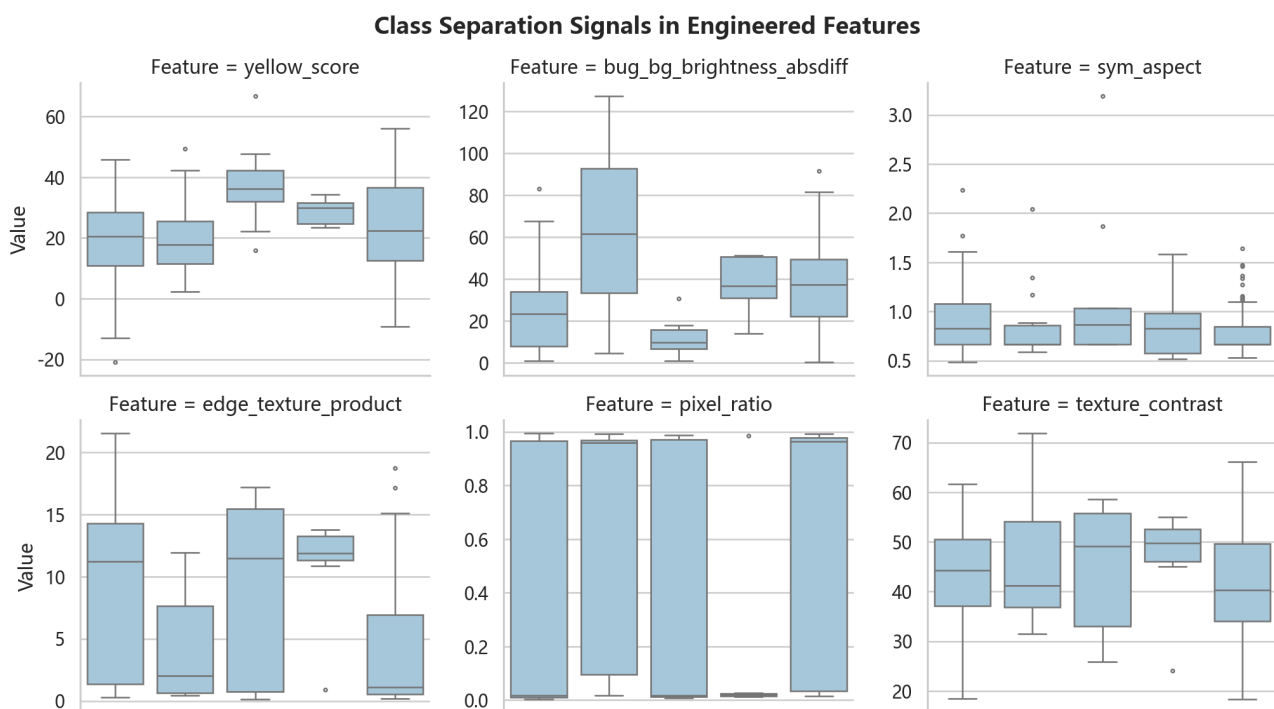


Figure 9. Interpretable engineered features by class.

Final Model Selection

Accuracy-oriented model: best_model_engineered.pkl. Model = Voting SVM+LDA+ET; strategy = none; CV accuracy = 0.8024 +/- 0.0462; CV macro F1 = 0.5979; holdout accuracy = 0.7600.

Balanced-class alternative: best_model_engineered_balanced.pkl. Model = SVM RBF C=10; strategy = hybrid_50; CV accuracy = 0.7136 +/- 0.0505; CV macro F1 = 0.6120; holdout macro F1 = 0.6389.

Prediction Pipeline

Extract the same 39 base features from each test image and mask.

Apply `add_engineered_features` to create the 80-feature representation.

Select the saved `feature_cols` from the pickle payload.

Call the saved model and write a CSV with columns ID and bug type.

Limitations and Conclusion

The saved final models predict five classes because two original labels have only one sample each. If the hidden test set contains those rare labels, an additional fallback rule is needed. Within the evaluated five-class setup, the engineered-feature soft voting ensemble is the strongest candidate.